WORKED SOLUTION

STEP III 2016, Question 4

Pure • Telescoping Series

STEP III 2016, Question 4

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/16-s3-q4

The key idea

The suggested difference of two fractions turns out to be a constant multiple of the general term, so the sum telescopes: almost everything cancels and only the first and last pieces survive. For part (ii), writing sech in terms of exponentials and pulling out a factor makes each term look exactly like the terms of part (i) with $x = \mathrm{e}^{-2y}$. The doubly infinite sum then folds back onto the one-sided sum, because sech is an even function.

Solution

Part (i)

Putting the suggested difference over a common denominator,

$$\frac{1}{1+x^r} - \frac{1}{1+x^{r+1}} = \frac{x^{r+1}-x^r}{(1+x^r)(1+x^{r+1})} = (x-1)\,\frac{x^r}{(1+x^r)(1+x^{r+1})}.$$

Since $|x| \neq 1$ we have $x \neq 1$, so we may divide by $x-1$. (The condition also guarantees $x \neq -1$, so no denominator $1+x^r$ vanishes: $x^r = -1$ would force $|x| = 1$.) Hence

$$\frac{x^r}{(1+x^r)(1+x^{r+1})} = \frac{1}{x-1}\left(\frac{1}{1+x^r} - \frac{1}{1+x^{r+1}}\right).$$

Summing from $r = 1$ to N, the second fraction of each bracket cancels the first fraction of the next, leaving

$$\begin{aligned}\sum_{r=1}^{N} \frac{x^r}{(1+x^r)(1+x^{r+1})} &= \frac{1}{x-1}\left(\frac{1}{1+x} - \frac{1}{1+x^{N+1}}\right)\\ &= \frac{x\,(x^N-1)}{(x-1)(1+x)(1+x^{N+1})},\end{aligned}$$

where the second form comes from combining the two fractions: the numerator is $x^{N+1} - x = x(x^N - 1)$.

Now suppose $|x| < 1$. Then $x^{N+1} \to 0$ as $N \to \infty$, so the partial sums tend to

$$\frac{1}{x-1}\left(\frac{1}{1+x} - 1\right) = \frac{1}{x-1} \cdot \frac{-x}{1+x} = \frac{x}{1-x^2},$$

which is the required value of the infinite sum.

Part (ii)

For any real t,

$$\operatorname{sech} t = \frac{2}{e^t + e^{-t}} = \frac{2e^{-t}}{1 + e^{-2t}}.$$

Let $x = e^{-2y}$; since $y > 0$ we have $0 < x < 1$. Then $\operatorname{sech}(ry) = 2e^{-ry}/(1 + x^r)$, and

$$\operatorname{sech}(ry)\operatorname{sech}((r+1)y) = \frac{4\,e^{-(2r+1)y}}{(1+x^r)(1+x^{r+1})} = 4e^{-y}\,\frac{x^r}{(1+x^r)(1+x^{r+1})},$$

because $e^{-2ry} = x^r$.

> **Ask yourself** Why split $e^{-(2r+1)y}$ as $e^{-y} \cdot e^{-2ry}$?

We want the numerator to be x^r, a power of the *same* x that appears in the denominator, so that part (i) applies term by term; the leftover factor e^{-y} does not depend on r and comes outside the sum.

Since $|x| < 1$, part (i) gives

$$\sum_{r=1}^{\infty} \operatorname{sech}(ry)\operatorname{sech}((r+1)y) = 4e^{-y}\,\frac{x}{1-x^2} = \frac{4e^{-y}}{x^{-1} - x} = \frac{4e^{-y}}{e^{2y} - e^{-2y}} = \frac{4e^{-y}}{2\sinh 2y},$$

which is $2e^{-y}\operatorname{cosech}(2y)$, as required. Call this sum S.

The doubly infinite sum

Split the sum over all integers r into three pieces: $r \geq 1$, $r \in \{-1, 0\}$, and $r \leq -2$.

> **Ask yourself** What do the terms with negative r look like?

For $r \leq -2$ put $s = -r - 1$, so $s \geq 1$ and $r + 1 = -s$, $r = -(s+1)$. As sech is even,

$$\operatorname{sech}(ry)\operatorname{sech}((r+1)y) = \operatorname{sech}((s+1)y)\operatorname{sech}(sy),$$

and as r runs over $-2, -3, \ldots$, s runs over $1, 2, \ldots$. So the terms with $r \leq -2$ also sum to S. (Both one-sided sums converge, so splitting the doubly infinite sum like this is legitimate.)

The two middle terms are

$$r = -1 : \ \operatorname{sech}(-y)\operatorname{sech} 0 = \operatorname{sech} y, \qquad r = 0 : \ \operatorname{sech} 0 \operatorname{sech} y = \operatorname{sech} y.$$

Therefore, using $\sinh 2y = 2\sinh y \cosh y$,

$$\sum_{r=-\infty}^{\infty} \operatorname{sech}(ry)\operatorname{sech}((r+1)y) = 2S + 2\operatorname{sech} y = \frac{2\mathrm{e}^{-y}}{\sinh y \cosh y} + \frac{2\sinh y}{\sinh y \cosh y}$$
$$= \frac{2(\mathrm{e}^{-y} + \sinh y)}{\sinh y \cosh y}.$$

But $\mathrm{e}^{-y} + \sinh y = \mathrm{e}^{-y} + \frac{1}{2}(\mathrm{e}^{y} - \mathrm{e}^{-y}) = \frac{1}{2}(\mathrm{e}^{y} + \mathrm{e}^{-y}) = \cosh y$, so the $\cosh y$ cancels and

$$\sum_{r=-\infty}^{\infty} \operatorname{sech}(ry)\operatorname{sech}((r+1)y) = \frac{2}{\sinh y} = 2\operatorname{cosech} y.$$

Checks

- With $N = 1$ the finite formula gives $\dfrac{x(x-1)}{(x-1)(1+x)(1+x^2)} = \dfrac{x}{(1+x)(1+x^2)}$, which is the single term $r = 1$.
- With $x = \frac{1}{2}$: the infinite sum should be $\frac{1/2}{3/4} = \frac{2}{3}$, and the partial sums do approach 0.6667.
- With $y = 1$: $2\mathrm{e}^{-1}\operatorname{cosech} 2 \approx 0.202864$ and $2\operatorname{cosech} 1 \approx 1.701836$, and both agree with direct summation of the series to many decimal places (confirmed by computer for many values of y).
- As $y \to \infty$ only the terms $r = -1, 0$ survive, each about $2\mathrm{e}^{-y}$, giving about $4\mathrm{e}^{-y}$; and $2\operatorname{cosech} y \approx 4\mathrm{e}^{-y}$ too.

Common slips

- Forgetting the factor $\frac{1}{x-1}$, or getting its sign wrong: the difference equals $(x-1)$ times the term, not $(1-x)$ times it.
- In the doubly infinite sum, doubling S and forgetting the two terms $r = -1$ and $r = 0$, which involve $\operatorname{sech} 0 = 1$ and are not part of either one-sided sum.

Beyond STEP

Rearranging and splitting an infinite sum is safe here because every term is positive (the series converges absolutely); for conditionally convergent series, Riemann's rearrangement theorem says reordering can change the sum to anything you like, a striking result met in a first analysis course. Sums of products like $\operatorname{sech}(ry)\operatorname{sech}((r+1)y)$ over all integers also appear in the theory of elliptic functions, where such bilateral series have beautiful closed forms. For the analysis of series, see Abbott, *Understanding Analysis*.

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